Muhammad Raees Azam

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LinkedIn | GitHub | Google for Developers | Portfolio | Linktree

OBJECTIVES

To pursue advanced studies in AI, machine learning, and robotics at a prestigious institution abroad, leveraging my expertise and achievements in computer engineering and STEAM Education. Working as Data scientist and AI/ML Engineer passionate about community impact, conducting workshops, and enriching academic and extracurricular environments with aim to give back to the community in the area of Education and Research under STEAM Curriculum to contribute to innovative research and community betterment.

EDUCATION

2026- In Progress

MSc Artificial Intelligence, Özyeğin University, Istanbul, Turkey

- **Semester**: 1st **| Thesis Domain:** Computer Vision, Federated Learning, Large Language Model and Embedded Systems
- **Core Courses :** Digital Image Processing, Large Language Models, Computer Vision, Data Science, Mechatronics Engineering

2020-2024

BS Computer Engineering, COMSATS University Islamabad, Abbottabad Campus, KPK, Pakistan

- **CGPA**:3.27 **| Status :** The Most Practical and Conceptual Student
- **Core Courses :** Electronics, Digital System Design, Embedded Systems, Artificial Intelligence, Control Systems, Machine Learning, Neural Networks, Digital Image Processing, Website Development

2017-2019

FSC Computer Science, Edwardes College Peshawar, KPK, Pakistan

- **Marks:**868/1100 **| Percentage:** 78.90% **| Core Courses :** Computer Science, Physics and Mathematics
- **Awarded with Double Gold Medal In Computer Science Department and Student Talent Award 2018**

PUBLICATIONS

1. M. Arshid , **M. R. Azam**, and Z. Mahmood, "A Comparative Study on Detection and Recognition of Non-Uniform License Plates," Big Data and Cognitive Computing, vol. 8, no. 11, 2024, pp. 1-20." **(Impact Factor: 3.7)**
2. **M. R. Azam**, Zeeshan Shafiq*, Adil Shah, Gul Muhammad Khan," Deep Learning-Based Precise Leaf Analysis for Segmentation and Deficiency Classification under Controlled Environment" **(Accepted)**
3. M. Arshid , **M. R. Azam**, and Shoaib Azmat*, "Generative AI Based Brain Tumar Segmentation of MRI Images" (In Preparation) Expected Submission Date: May 2025
4. Mehak Arshid , **Muhammad Raees Azam*** ,Muhammad Danyal and Sardar Muhammad Gulfam*, "Text-Independent Speaker Recognition and Audio Integrity Verification in Next-Generation Communication Networks using MFCCs and Machine Learning" (Accepted)
5. Syeda Rida Zainab Kazmi , **Muhammad Raees Azam** , and Sardar Muhammad Gulfam ,*"Robust Sign Language Generation and Recognition: Real-Time Translation with Error Handling in Sign Generation" " (In Preparation) Expected Submission Date: April 2025

Professional Appointments

TÜBİTAK Project, Istanbul, Türkiye.

Researcher

Oct 2025-Till Now

- Conducting research and development on Smart Weeding Systems using fiber laser technology for autonomous weed detection and removal in controlled greenhouse environments.
- Designing and training custom deep learning models for weed localization and detection, utilizing advanced computer vision and image segmentation techniques.
- Integrating real-time AI inference pipelines on embedded edge devices such as NVIDIA Jetson Nano and Raspberry Pi for on-site weed identification and laser activation.
- Developing sensor-based feedback and control systems to ensure precision targeting and safe laser operation for selective weeding.
- Optimizing model performance for low-latency and energy-efficient operation, ensuring scalability for agricultural automation applications.
- Collaborating with multidisciplinary teams on hardware-software integration, data collection, annotation, and performance evaluation in real-world greenhouse setups.
- Exploring innovative approaches in AI-powered agricultural robotics, focusing on sustainable, chemical-free weed management through intelligent automation.

  Tools & Technologies: Python, TensorFlow, PyTorch, OpenCV, Jetson Nano, Raspberry Pi, Laser Control Systems, Edge AI Deployment, Image Segmentation, Object Detection, Embedded Systems, IoT, Computer Vision, Deep Learning.

Neuralogics, United States of America.

Artificial Intelligence Engineer

Aug 2025-Till Now

- Working on advanced projects involving Vision-Language Models (VLMs), integrating computer vision and natural language understanding for real-world automation and analytics tasks.
- Developing intelligent data pipelines for large-scale data scraping, web crawling, and data cleaning, ensuring high-quality datasets for model training and evaluation.
- Designing and implementing automation tools to streamline data collection, preprocessing, and AI model deployment workflows.
- Collaborating on multimodal AI systems that combine text, image, and structured data to enable context-aware decision-making and generative capabilities.
- Building and fine-tuning LLM-powered automation agents for web-based task execution, content extraction, and process optimization.
- Integrating and optimizing custom APIs for AI-driven data acquisition and real-time analytics across diverse domains..

  Tools & Technologies: Python, PyTorch, TensorFlow, OpenAI/Gemini APIs, LangChain, Selenium, BeautifulSoup, Scrapy, Pandas, NumPy, Streamlit, Flask, Git, Google Cloud Platform, REST APIs, Vision-Language Models (VLMs), Automation Frameworks.

CCRIPT Agency, Toronto, Canada.

Artificial Intelligence Engineer Project Lead

July 2025-Till Now

- Leading international client projects in Artificial Intelligence, focusing on delivering scalable, production-ready AI solutions.
- Developing and deploying advanced AI systems including Document AI pipelines, automation workflows (n8n), Agentic AI systems, Retrieval-Augmented Generation (RAG) models, and ReAct-based agents.
- Designing and implementing end-to-end solutions from requirement gathering, data preprocessing, and model development to deployment and monitoring.
- Collaborating with cross-functional teams to integrate AI into diverse business domains, ensuring efficiency, scalability, and real-world impact.
- Driving innovation in LLMs, NLP, computer vision, and generative AI applications tailored to client needs.
- Conducting performance optimization, fine-tuning, and evaluation of AI models to ensure reliability and accuracy.
- Managing project timelines, stakeholder communication, and technical documentation for smooth execution.
- Exploring emerging AI tools and frameworks to continuously improve project outcomes and maintain a competitive edge.

  Tools & Technologies: Python, Scikit-learn, TensorFlow, PyTorch, OpenAI/Gemini APIs, LangChain, Streamlit, Flask, Git, n8n, RAG, ReAct Agents, Document AI, Google Cloud Platforms.

Arfa Karim Technology Incubator, Peshawar, Pakistan.

Artificial Intelligence Trainer

March 2025-September 2025

- Leading 3-month, project-based training programs in Generative AI, Data Science, and Huawei HCDI AI, mentoring 75+ students from diverse backgrounds.
- Delivering hands-on sessions covering Python programming, EDA, ML algorithms, LLMs (OpenAI, Gemini), LangChain, NLP, CV, and model deployment.
- Guiding students in building real-world AI solutions including chatbots, resume screeners, fraud detection, transcription tools, and object detection apps.
- Creating and managing interactive course content, coding assignments, and capstone projects tailored to different learning levels.
- Providing continuous mentorship and personalized feedback to enhance students' technical proficiency and problem-solving skills.

  Tools and Technologies: Python, Scikit-learn, TensorFlow, OpenAI/Gemini APIs, LangChain, Streamlit, Google Colab, Flask, IoT kits, Git.

BanoQabil IT Training Program, Peshawar, Pakistan.

Artificial Intelligence & Machine Learning Trainer

August 2025-September 2025

- Currently serving as an AI/ML Trainer for the BanoQabil IT Training Program, a national initiative aimed at equipping youth with industry-relevant IT skills.
- Delivering comprehensive training on Artificial Intelligence and Machine Learning tools and techniques, covering Python programming, ML algorithms, deep learning frameworks, NLP, computer vision, and generative AI.
- Designing and conducting hands-on workshops, coding labs, and real-world projects to bridge the gap between academic knowledge and practical industry application.
- Mentoring students in developing end-to-end AI solutions, including intelligent chatbots, document processing pipelines, fraud detection systems, and RAG-based applications.
- Providing continuous guidance, technical feedback, and career mentorship, ensuring students gain problem-solving ability, research exposure, and workplace readiness.
- Collaborating with Alkhidmat Foundation and EncoderBytes to align curriculum with evolving AI/ML industry demands, ensuring graduates are equipped with cutting-edge skills and tools.

Stixor Technologies, Islamabad, Pakistan.

AI/ML Developer

April 2025-July 2025

- Designed and developed advanced AI agent frameworks to automate complex workflows and support intelligent decision-making.
- Built generative AI models and multi-agent systems leveraging Large Language Models (LLMs) such as OpenAI and Gemini APIs.
- Created intelligent, customizable conversational AI agents capable of autonomous and human-assisted operations.
- Integrated APIs, databases, and third-party tools into cohesive AI pipelines for optimized data access, retrieval, and task execution.

- Implemented agent orchestration to enhance tool selection, coordination, and dynamic reasoning.
- Developed machine learning models for fraud detection to identify and prevent suspicious transactions for JazzCash using XGBoost, LGBM, and ensemble learning techniques.
- Contributed to cross-functional teams to design and deliver scalable, production-grade AI systems aligned with real-world business problems.
- Actively participated in AI research and experimentation to evaluate emerging Agentic AI, custom deep learning models, and multimodal systems.

Key Projects:

- **Fraudulent Transaction Detection System** – Developed an ML-based fraud detection engine for JazzCash, using XGBoost and LGBM for predictive analytics.
- **Smart Toll Collection System** – Implemented a YOLOv11/YOLOv12 based number plate recognition system integrated with automated billing for NHA.
- **AI-Powered Transcription Reimbursement System** – Built a generative AI system for automated medical transcription and verification for FCCL Pharmacy Claims.
- **Import/Export Duty Chatbot** – Developed an intelligent chatbot using Google Search APIs, web scraping, and LLMs for real-time customs duty information at AlMakhdoom Logistics.

Tools and Technologies: Python, OpenAI API, Gemini API, Semantic Kernel, YOLOv11/v12, XGBoost, LightGBM, LangChain, Google Search APIs, Custom LLMs, REST APIs, Flask, GitHub.

Infaque Inc, Toronto, Canada.

Embedded Engineer

Feb 2025-Till Now

- Developed and deployed IoT-enabled software systems on Raspberry Pi for real-time agricultural applications.
- Built an interactive desktop application for a Weight Cell Management System, enabling automated vegetable identification using image and weight data via integrated AI models.
- Implemented real-time data acquisition and transmission (weight, type, timestamp) to a central database for monitoring and analytics.
- Ensured seamless integration of hardware components, AI models, and software platforms through cross-functional collaboration.
- Currently working on a Smart Agricultural AI-Based Drone System for precision spraying and crop health monitoring, using deep learning models and aerial imaging.

Tools and Technologies: Python, Raspberry Pi, OpenCV, CNN, Firebase, IoT Sensors, Image Classification, DroneTech

Github Repo:

National Center of Artificial Intelligence, UET Peshawar

Data Engineer

Aug 2024 – April 2025

- Developing AI algorithms and models under a PKR 140+ million HEC-funded research project focused on smart agriculture and precision farming.
- Designed and optimized multiple machine learning and deep learning models, improving accuracy and reducing model latency in real-time deployment environments.
- Analyzed large-scale agricultural datasets to enhance model training and testing efficiency, contributing to the creation of robust, scalable AI systems.
- Built and deployed 5 smart incubators for controlled agricultural environments, achieving 2× crop yield and reducing growth time by 1.5× compared to traditional field methods.
- Collaborated with cross-functional teams of agronomists, AI researchers, and hardware engineers to align technical solutions with practical field requirements.
- Successfully deployed AI models into production with ongoing monitoring, performance evaluation, and iterative tuning.
- First author of the research paper titled:
- "Deep Learning-Based Precise Leaf Analysis for Segmentation and Deficiency Classification Under Controlled Environment"
- Published in: Artificial Intelligence Applications and Innovations – AIAI 2025, IFIP WG 12.5 International Workshops, Chapter No: 24, DOI: 10.1007/978-3-031-97313-0_24

Tools and Technologies: Python, TensorFlow, Keras, CNNs, FCN-8s, OpenCV, Yolo, ResNet50, FCN, UNET , Drone Imaging, IoT Sensors, Raspberry Pi, Jetson Nano, Nodemcu, SQL, Git, Google Colab, Kaggle, LaTeX.

Department of Computer Engineering, COMSATS University Islamabad, Abbottabad Campus, Pakistan

Research Fellow

July 2024-Dec 2024

- Conducted deep analysis of existing License Plate Detection and Recognition (LPDR) methods under unconstrained environments, focusing on challenges like varying illuminations, nonstandard templates, and multilingual fonts.
- Collected and annotated a custom dataset of Pakistani license plates with diverse styles and environmental conditions, enabling real-world experimentation.
- Developed and evaluated multiple deep learning models for LPDR using YOLO, Faster-RCNN, EAST, CRNN, and CA-CenterNet, achieving 98.41% average detection accuracy and up to 98.96% recognition accuracy.
- Optimized model performance using GPU-accelerated training with PyTorch and TensorFlow, significantly reducing training time and enhancing model precision.
- Contributed to a research publication (Impact Factor 3.7) titled "A Comparative Study on License Plate Detection and Recognition Algorithms in Unconstrained Environments" by drafting sections, designing experiments, and developing technical rebuttals during peer review.

- The study concluded that E2E detection methods and CA-CenterNet-based recognition are more accurate and computationally efficient, especially in developing-country conditions like Pakistan.
  Tools and Technologies: Python, PyTorch, TensorFlow, OpenCV, YOLO, CRNN, CA-CenterNet, Google Colab, CUDA

National Center of Robotics and Automation, UET Peshawar

Robotics Engineer Intern

Aug 2022 – Sep 2022

- Supervised STEM Robotic Kit Manufacturing Project for providing better learning opportunities,Engaged with Agricultural UGV for spraying the plants Robots,Teaching Students in the STEAM Domains.

National STEM School, Maker's Summer Camp, Pakistan Innovation Foundation

Robotics Teacher's Assistant

July 2019

- Assisted lead instructors in teaching and supervising robotics and electronics projects for high school and undergraduate students, fostering hands-on learning in STEM fields.
- Guided students in building and programming robotics prototypes, providing mentorship in problem-solving, electronics design, and mechanical assembly.
- Actively supported curriculum delivery by preparing lab materials, setting up experiments, and ensuring safety compliance in all practical sessions.
- Collaborated with faculty and peers to provide personalized feedback and technical support, helping students strengthen their understanding of robotics concepts.
- Managed and coordinated the Makers Summer Camp held at the College of Electrical & Mechanical Engineering, NUST, overseeing logistics, scheduling, and smooth execution of daily activities.
- Encouraged teamwork and innovation among participants, cultivating an engaging environment that inspired creativity, critical thinking, and interest in robotics.

Final Year Project

AgriTech - A Robust Artificial Intelligence Based Crop Monitoring System for Pakistani Agricultural Land

Addressed the limitations of traditional crop monitoring methods by creating a system that analyzes rice crop health using RGB aerial images. Key features include

- **Aerial Imagery:** Used RGB images captured from aerial drones for crop analysis and Targeted Spraying.
- **Image Processing & AI:** Applied image processing techniques along with FCN-8s for segmentation and CNN for disease classification.
- **Mobile Application:** Developed a real-time classification mobile application using Flutter to provide immediate insights on crop health issues.
- **Funding Received:** PKR: 350,000 Rs
- **Tools/ Techniques:** Google COLAB, Mission Planner, Debian, Raspberrypi, Arduipilot, Python, CNN, FCN 8s, Segmentation and Classification, Kaggle, Flutter, L298N, GPS, Keras, Tensorflow.
- **Project Achievements:** Top 100 Finalist Globally in Google Solution Challenge 2024, Fully Funded by Pakistan Engineering Council, 2nd Prize winner of KP Capstone Expo 2024 by PEC, Finalist in Google Cloud Startup Competition Pakistan, Funding winner of NGIRI-Ignite, Shortlisted for P@sha ICT Award 2024.

SYNOPSIS OF ACHIEVEMENTS

Academic Achievements

- 2nd Prize Winner at **KP Capstone Expo 2024** by Pakistan Engineering Council within 100+ top shortlisted Project around the Province.
- My Final Year Project (FYP) has been fully funded by **Pakistan Engineering Council** for complete project duration from years 2023-2024 worth of 0.2 Million PKR and Ignite ICT Funds worth 98,000 PKR.
- I secured the 2nd position in the **TechFestFall21 Project Exhibition Competition**, which took place at the COMSATS University Islamabad, Abbottabad Campus and featured a total of 45 contestants.
- Selected as a Teacher's Assistant in **PIF's National STEM School** in NUST CEME 2019.
- Shield of Appreciation at **Student Talent Expo** Continuously from 2016, 2017, 2018 and 2018.
- I was the Best Graduate and got **Gold Medal** from Edwardes College Peshawar during my Intermediate studies along with full scholarship.
- Appreciation Certificate at **Pre-STEM Exhibition by Edwardes STEM Societ**y in Edwardes College Peshawar 2018 by Presenting IoT Project of Home Automation.

Co-Curricular Achievements

- Trained 500+ Students in AISeekho Season 6 across Pakistan through Google for Developers Plateform.
- Invited as **Guest Speaker Trainer** at AISeekho Season 6 by Google for Developers Team, Google Developers Group Peshawar.
- Invited as **Guest Speaker** at GDSC Global Graduation Ceremony 2024 by Google for Developers.
- Declared as Top 100 Finalist Globally of **Google Global Solution Challenge 2024.**
- Shield of Resource Person at GCT Abbottabad for **"Arduino 101 Workshop"** in 2023.
- Trainer on multiple Workshops with IET On Campus, IEEE and GDSCs Chapters
- Selected as a Google Developer Student Club Lead by **Google for Developers** for 2023 – 2024 and managed 500+ Members.
- 3rd Prize Winner at **Multiomics Creative Competition** in Logo Designing Competition 2020.
- Winner of **NASA Space App Challenge 2019** at CECOS University by Team RaheQamar.
- Top 25 Finalist in **Pakistan Startup Cup 2018** from Peshawar, KPK for presenting MeatInspector
- 3rd Prize winner in **Startup Weekend Peshawar 2017** at Basecamp, Peshawar2.0 for presenting MeatInspector